

# Analgesics

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# Learning Outcomes

By the end of this lecture, students should be able to:

Describe the mechanism of action of opioid analgesics

Outline the side effects of opioid analgesics

Describe the mechanism of action of Non-steroidal anti-inflammatory drugs

List therapeutic uses of Non-steroidal anti-inflammatory drugs in dentistry

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# What are analgesics

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Analgesics are the drugs (natural or synthetic origin) which relieve pain by acting on CNS or peripheral pain mechanism without causing loss of consciousness.

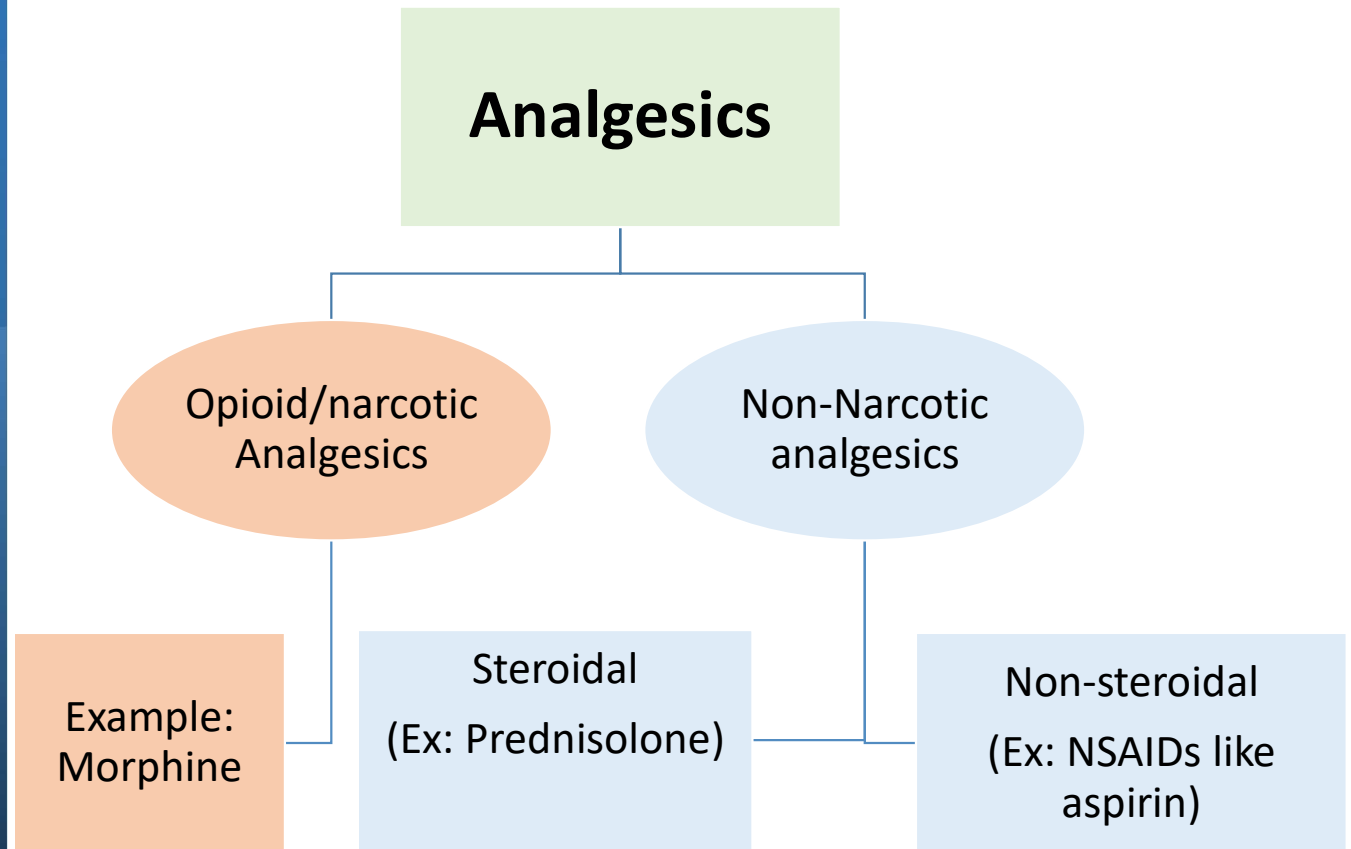
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Dental pain is the initial sign of any dental disease which is usually acute in nature.

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In dentistry, non-steroidal anti-inflammatory drugs (NSAID's) are commonly employed for dental pain and inflammation which may be due to caries, abscess, tooth extraction or after any dental procedures.

# Classification





# OPIOID/NARCOTIC ANALGESICS

# History of Opioids

Opium is  
extracted from  
poppy seeds  
(*Papaver  
somniferum*)

Used for thousands of years to  
produce:

- Euphoria
- Analgesia
- Sedation
- Relief from diarrhea
- Cough suppression
- Morphine was isolated from opium in the early 1800's and since then has been the most effective treatment for severe pain.

# Mechanism of action

By acting on different receptors namely mu ( $\mu$ ), kappa ( $\kappa$ ) and delta ( $\delta$ ).

Analgesic, respiratory depression as well as euphoria produced by morphine result mainly from action at mu receptors.

Most act on mu receptors.

Delta and kappa receptors can also contribute to analgesia, particularly at spinal level.

# Pharmacological Effects

## Sedation and anxiolytics

- Drowsiness and lethargy
- Apathy(lack of interest, enthusiasm, or concern)
- Cognitive impairment
- Sense of tranquility (Calmness)

## Depression of respiration

- Main cause of death from opioid overdose
- Combination of opioids and alcohol is especially dangerous

## Cough suppression

- Opioids suppress the “cough center” in the brain.



## Pharmacological effects cont'd.

### Nausea and vomiting

- Stimulation of receptors in an area of the medulla called the chemoreceptor trigger zone causes nausea and vomiting
- Unpleasant side effect, but not life threatening

### Gastrointestinal symptoms

- Opioids relieve diarrhea as a result of their direct actions on the intestines

### Other effects

- Opioids can release histamines causing itching or more severe allergic reactions including bronchoconstriction.



How to  
overcome  
toxicity?

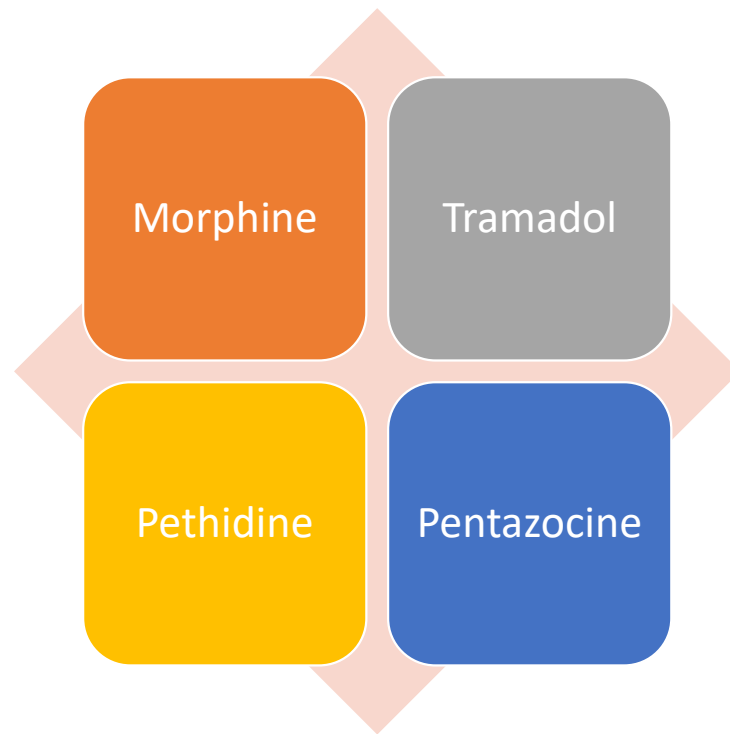
## Acute Overdose/Toxicity (Morphine Poisoning) Symptoms

- Respiratory depression, Miosis, cyanosis, reduced body temperature, urinary retention, hypotension, pulmonary edema and coma.

## Treatment

- Maintenance of airway and oxygen inhalation.
- Maintenance of BP.
- Specific antagonists: **Naloxone**

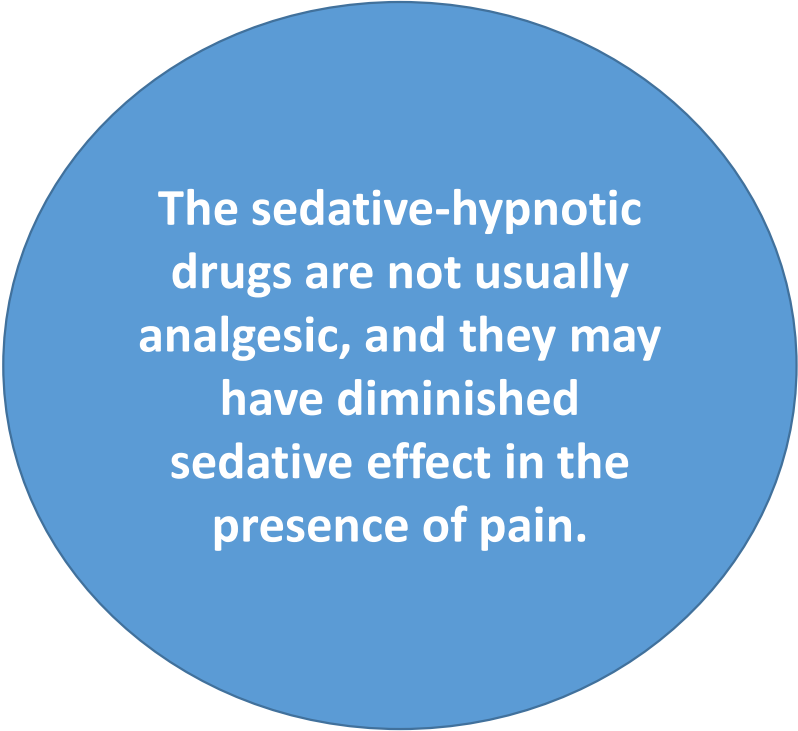
# Opioid drugs



# Uses of Morphine

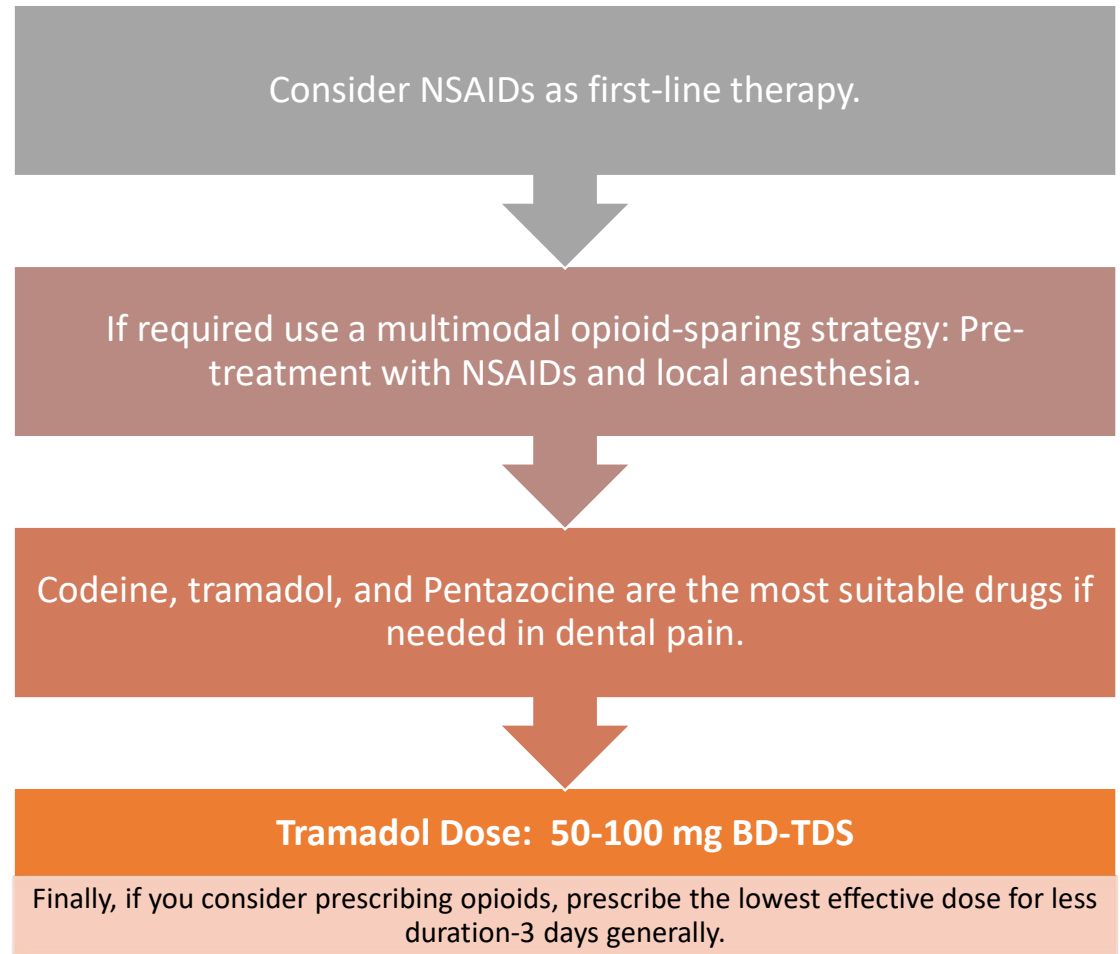
**Analgesia:** Opioids induce sleep, and in clinical situations when pain is present and sleep is necessary, opiates may be used to supplement the sleep-inducing properties of benzodiazepines, such as temazepam.

- Mainly used in severe pain e.g. myocardial infarction, renal and biliary colic, long bone fractures and burns.
- Treatment of diarrhea, Relief of cough, Treatment of acute pulmonary edema
- Use of opioids in dentistry is minimal



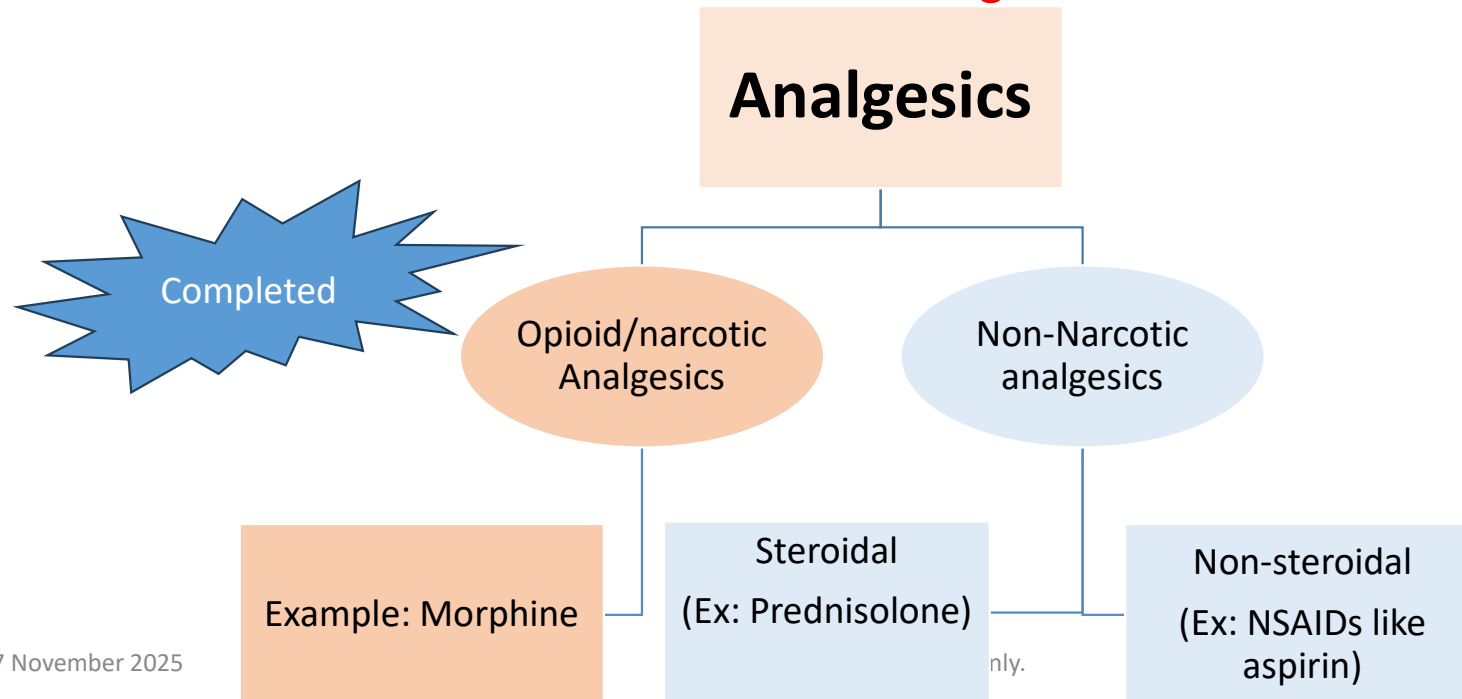
**The sedative-hypnotic drugs are not usually analgesic, and they may have diminished sedative effect in the presence of pain.**

## Usage in Dentistry



# Steroidal and Non-steroidal anti-inflammatory drugs(NSAIDs)

Non- Narcotic Analgesics

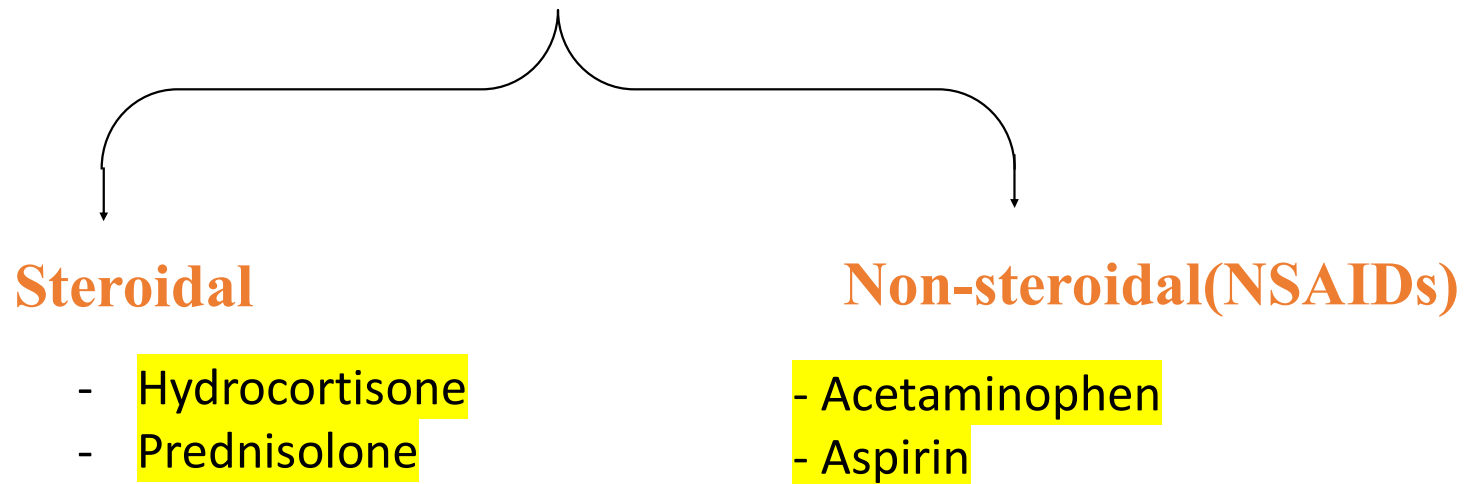


# Inflammatory mediators

- Histamine
- 5-HT (serotonin)
- Bradykinin
- **Prostaglandins** (eg PGE2 )
- Interleukines
- Substance P
- Nitrous oxide



# Anti-inflammatory Drugs





# Steroidal anti-inflammatory drugs

# Mechanism of action



They inhibit enzyme phospholipase A2



So, that the arachidonic acid will not be converted to prostaglandins.

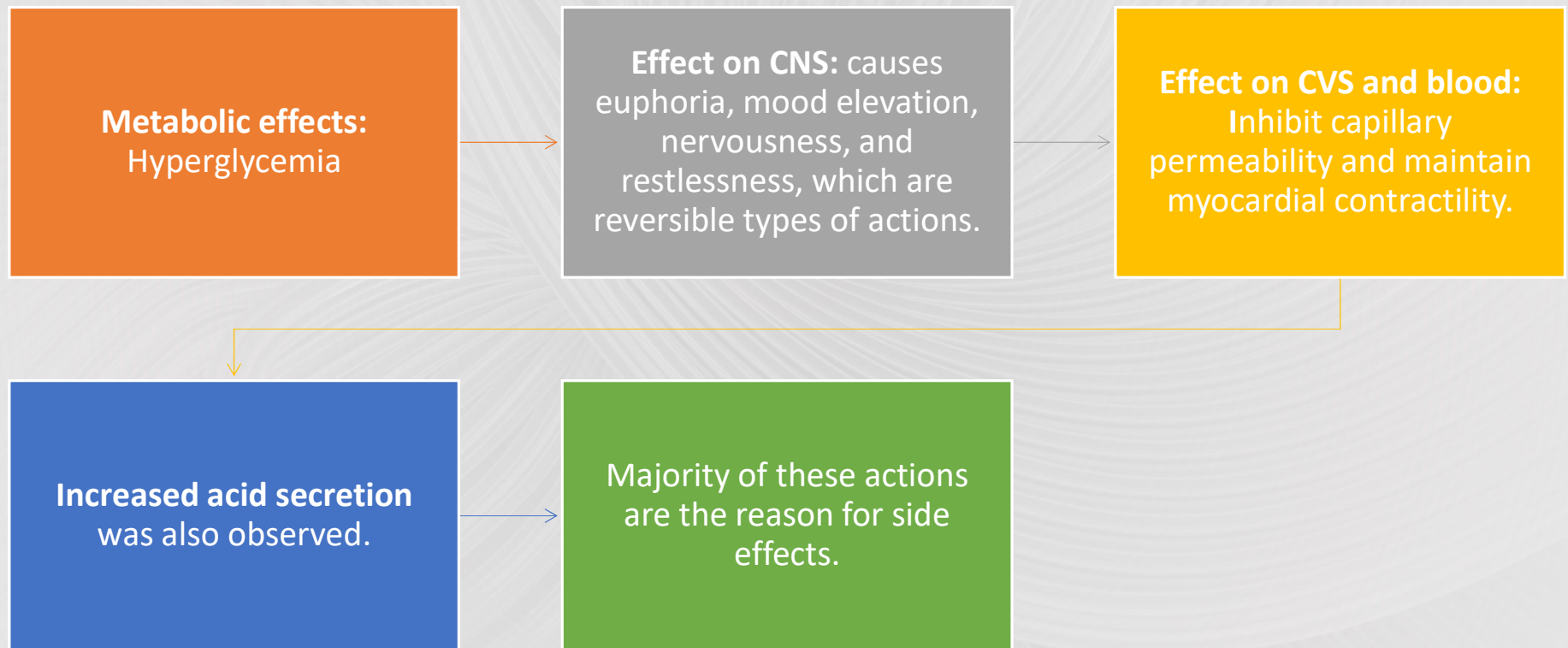


Glucocorticoids **suppress all types of inflammation and allergic** reactions.



They suppress the edema, capillary dilatation, migration of leukocytes, and capillary permeability in the inflamed area.

## Other Pharmacological actions of glucocorticoids(Hydrocortisone, Prednisolone)



# Non- Narcotic analgesics(NSAIDs)

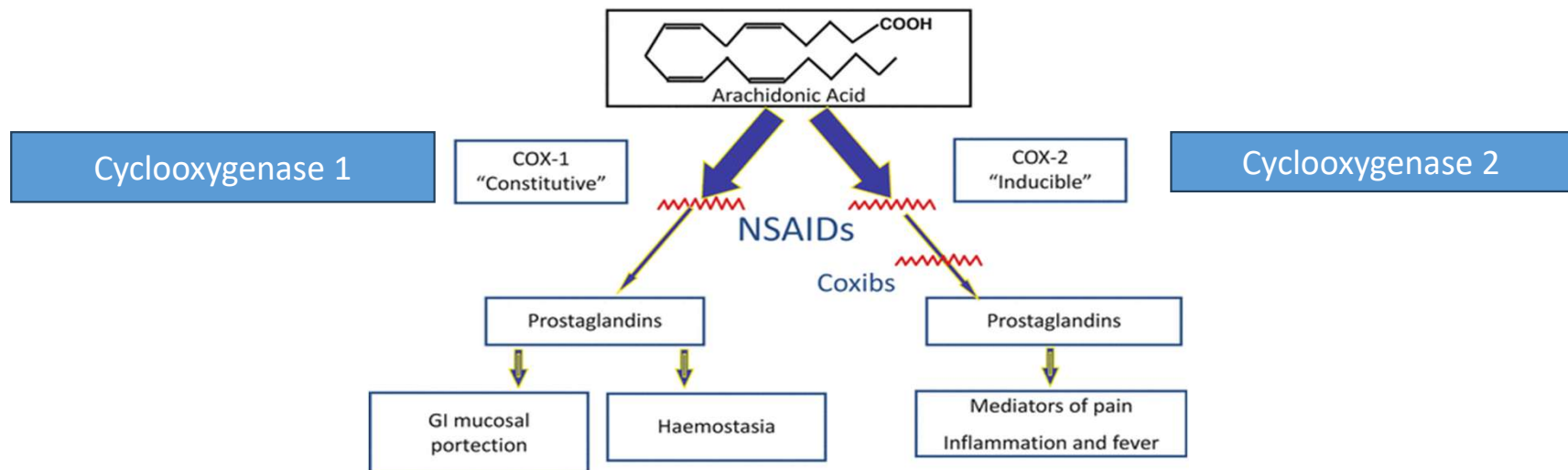
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# Introduction

- **NSAIDs have three major types of effects:**
- Anti-inflammatory effects
- Analgesic effect
- Antipyretic effect
- **Mechanism:** Inhibition of **cyclooxygenase** and thus inhibition of the production of prostaglandins and thromboxanes.

# Pathogenesis of pain

## Mechanisms of action of NSAIDs



# Effects of COX Inhibition by NSAIDs

COX-1

Causes  
Gastric ulcers

Causes  
Bleeding

Causes Acute  
renal failure

So inhibition of COX 1 causes side effects

COX-2

Reduce  
inflammation

Reduce pain

Reduce  
fever

So, inhibition of COX 2 alone is good

# Adverse effects



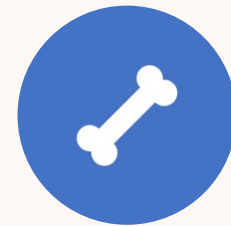
NAUSEA, VOMITING,  
GASTRIC ULCERS IN  
CHRONIC USERS



SKIN REACTIONS.



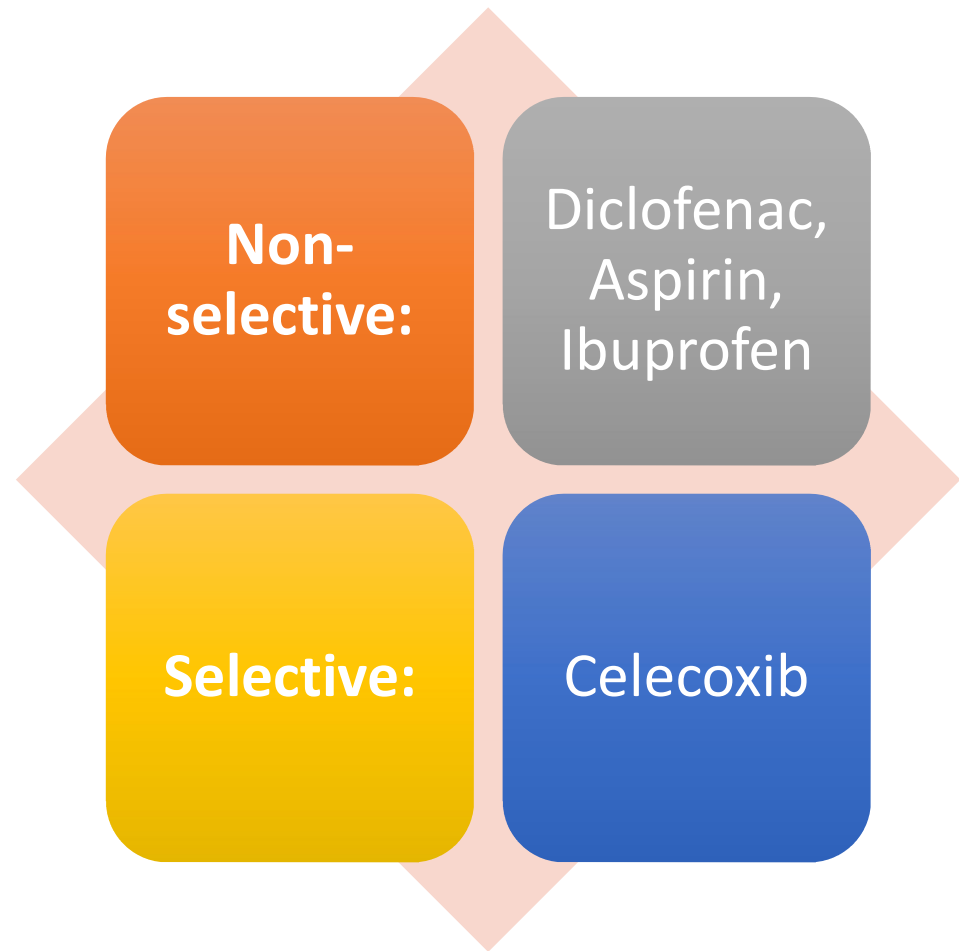
ANALGESIC-  
ASSOCIATED  
NEPHROPATHY



LIVER DISORDERS,  
BONE MARROW  
DEPRESSION, AND  
BRONCHOSPASM IN  
ASTHMATICS.



# Classification of NSAIDs



# Asprin

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**Analgesic action:** Salicylates relieve pain by both central and peripheral action.

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Unlike morphine, they **do not produce sedation** and there is no drug tolerance or dependence.

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Also, works as an antiplatelet drug

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Relieve mild to moderate pain, and reduce inflammation & and fever.

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## **FOR ANALGESIA OR ANTIPYRESIS IN ADULTS:**

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500 mg every 4 hrs. or 1000 mg every 6 hrs.

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The maximum daily dose is 4000 mg

# Aspirin-Side effects

**Salicylism:** It is a syndrome consisting of tinnitus (a high-pitched buzzing noise in the ears), vertigo, decreased hearing, and sometimes nausea and vomiting.



Other unwanted systemic effects like ***Reye's syndrome***, a rare disorder in children. (swelling in the liver and brain).

# Paracetamol

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No significant anti-inflammatory effect, but used for its mild analgesic effect.

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Well-absorbed and without GIT irritation.

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Analgesia – both centrally and peripherally.

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**Serious disadvantage:** at high doses, severe hepatotoxicity results.

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Dose: 500-1000 mg three times daily (as needed).

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The maximum daily dose is 4000 mg for no longer than 10 days for pain or 3 days for fever.

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**For children > 2 years:** 10 mg/kg/dose every 4-6 hrs (as needed).

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The maximum daily dose is 1600 mg for no longer than 5 days for pain, 3 days for fever, or 2 days for sore throat.

# Precautions

Patients with *active alcoholism*, *hepatic disease*, or *viral hepatitis* are at risk from chronic daily administration of acetaminophen.



## Significant interactions.

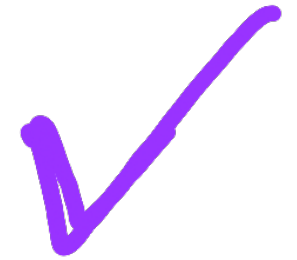
Acetaminophen may competitively inhibit the metabolism of **zidovudine**, resulting in potentiation of zidovudine or acetaminophen toxicity.

## Precautions for NSAID usage

1. Contraindicated in bleeding disorders, peptic ulcers, pregnancy (like Aspirin)
2. GI Adverse effects **Ibuprofen** causes fewer GI disturbances and bleeding events than aspirin.
3. Renal toxicity during chronic administration is a significant concern, and may occur in the form of nephrotic syndrome, hyperkalemia.
4. Cardiovascular toxicity of NSAIDs. Although the risk is low, **ibuprofen** use is more risky than naproxen.
5. Hypersensitivity can occur with NSAID.

Examples of Anticipated Post procedural Pain Levels According to Dental Intervention  
Adapted from Hersh et al. 2012

| Intervention                                                                                                                         | Anticipated Postprocedural Pain |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Frenectomy<br>Gingivectomy<br>Routine endodontics<br>Scaling/root planing<br>Simple extraction<br>Subgingival restorative procedures | Mild                            |
| Implant surgery<br>Quadrant periodontal flap surgery with bony recontouring<br>Surgical endodontics<br>Surgical extraction           | Moderate                        |
| Complex implant<br>Partial or full bony impaction surgery<br>Periodontal surgery                                                     | Severe                          |



## Analgesic Use According to Pain Level

| Anticipated Pain Level | Analgesic Recommendation                                                                                                                                                                                                      |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mild                   | Single-agent ibuprofen 200 to 400 mg as needed                                                                                                                                                                                |
| Mild to Moderate       | Single-agent ibuprofen 400 to 600 mg fixed interval every 6 hours for 24 hours<br><i>then</i><br>Single-agent ibuprofen 400 mg as needed for pain every 4 to 6 hours                                                          |
| Moderate to Severe     | Ibuprofen 400 to 600 mg plus acetaminophen 500 mg fixed interval every 6 hours for 24 hours<br><i>then</i><br>Ibuprofen 400 mg plus acetaminophen 500 mg as needed for pain every 6 hours                                     |
| Severe                 | Ibuprofen 400 to 600 mg plus acetaminophen 650 mg with hydrocodone 10 mg fixed interval every 6 hours for 24 to 48 hours<br><i>then</i><br>Ibuprofen 400 to 600 mg plus acetaminophen 500 mg as needed for pain every 6 hours |



Thank you

